

Week 5 Feasibility Memo

AI-Assisted Triage: Data Exploration

Prepared by: Shari Oliver

Programme: CariSurg MedTech Pathways Healthcare AI Programme

Project: AI-Assisted Emergency Department Triage Support

Dataset: De-identified ED arrivals dataset

A. One-Sentence Verdict

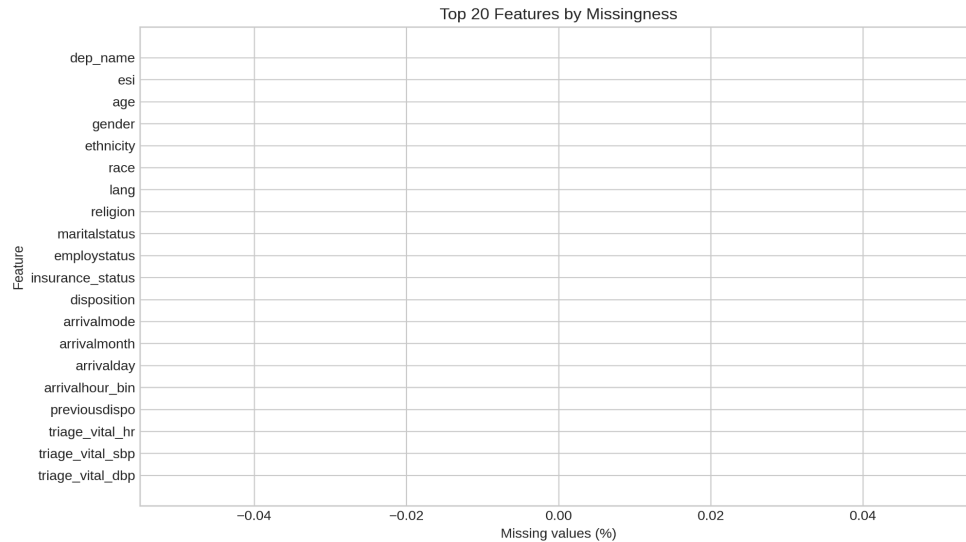
The dataset is suitable for continued exploration and Week 6 baseline triage modelling, but only with important caveats around missingness, ESI class balance, clinical plausibility, potential data leakage and single-site generalisability.

B. Dataset Summary

The dataset contains de-identified emergency department arrival records from Mercer General Hospital, which include structured clinical features, demographic variables, triage vital signs, Emergency Severity Index (ESI) triage levels and a large set of chief complaint indicators. The main modeling target for this stage of the project is ESI, which represents the patient's triage acuity level. The dataset is clinically useful because it includes variables that are commonly available during emergency department triage. These include age, arrival information, respiratory rate, oxygen saturation, blood pressure, heart rate, temperature, glucose and presenting complaints. These features are relevant because triage decisions are usually based on a combination of physiological status, presenting symptoms and clinical concern.

The Week 5 exploration focused on profiling the dataset before modeling. This included checking missingness, reviewing data types, exploring the ESI target distribution, assessing vital sign distributions, identifying outliers and clinically implausible values, reviewing fairness-sensitive variables and checking simple relationships between selected clinical features and ESI. ***NB: Additional plots supporting ESI distribution, race/ethnicity review, vital sign outliers and feature correlations are included in the Week 5 plots folder.***

Plot 1: Top 25 Features by Missingness



This plot supports the data-quality review by showing which fields have the highest levels of missingness. Missingness matters clinically because an AI triage tool can only make safe recommendations using information that is actually available at the time of triage.

C. Top 3 Quality Concerns

1. Missingness in key clinical fields

The first major concern is missingness in clinical fields. Missing values are especially important in triage because vital signs and presenting complaints directly affect the assessment of patient risk. If important fields such as oxygen saturation, respiratory rate, blood pressure, heart rate or ESI are missing, the model may not have enough information to support a safe prediction. Missing ESI labels are the most serious concern for supervised modeling because ESI is the target variable. Any record without a valid ESI label cannot be used to train a model that predicts triage level. Missing vital signs also need careful handling. These values should not be filled in automatically without clinical justification because missingness may reflect real emergency department workflow issues, such as delayed documentation, high workload or incomplete assessment at the time of triage.

2. Potential imbalance in ESI levels

The second concern is a possible imbalance in ESI triage categories. If one ESI level is much more common than the others, a baseline model may appear accurate while still performing poorly for less common but clinically important acuity levels. This is especially important for high-acuity patients because missing a seriously ill patient would be a major safety issue. For Week 6, model performance should therefore not be judged by overall accuracy alone. The

model should be reviewed across ESI levels to ensure that it is not simply learning to predict the most common category.

3. Clinical plausibility, data leakage and single-site limitations

The third concern is that some variables may require review for clinical plausibility and modeling appropriateness. Extreme vital sign values should not be removed automatically because they may represent true clinical deterioration. However, values outside plausible clinical ranges should be flagged because they may reflect documentation errors, unit errors or data entry problems. There is also a risk of data leakage if the model uses variables that would not be known at the time of triage, such as disposition or post-triage outcome variables. These should be excluded from Week 6 baseline modeling. Finally, the dataset reflects one hospital context, so any findings should be treated as site-specific until externally validated.

D. Top 3 Reasons to Proceed

1. The dataset contains clinically relevant triage features

The dataset includes several variables that clinicians would reasonably expect to influence triage decisions, including vital signs, age, arrival mode and chief complaint indicators. These are meaningful starting points for a baseline triage model.

2. ESI is a meaningful target for baseline modeling.

ESI is an appropriate starting target because it reflects triage acuity and is directly related to patient prioritization. Although ESI may reflect local documentation habits and workflow, it is still a practical target for initial modeling.

3. The dataset is rich enough for exploratory modeling.

The dataset contains many structured and chief complaint features, which makes it suitable for exploratory modeling. Simple correlation checks and clinical reasoning can be used together to guide the initial feature shortlist for Week 6.

Initial Top-10 Feature Shortlist

The following features are recommended for initial consideration in Week 6 baseline modelling. This shortlist is based on clinical relevance, expected relationship with acuity and early exploration of the dataset.

Rank	Feature	Reason for Inclusion
1	Respiratory rate	An abnormal respiratory rate can indicate respiratory distress, sepsis, shock, pain or deterioration.
2	Oxygen saturation	Low oxygen saturation directly reflects impaired oxygenation and may indicate urgent respiratory or cardiovascular risk.
3	Systolic blood pressure	Very low or very high blood pressure may suggest shock, bleeding, sepsis, stroke risk or cardiovascular instability.
4	Heart rate	Tachycardia or bradycardia may reflect pain, fever, dehydration, arrhythmia, shock or deterioration.
5	Temperature	Fever or hypothermia may indicate infection, sepsis or systemic illness.
6	Glucose	Abnormal glucose may indicate diabetic emergencies or an increased risk of altered mental status.
7	Age	Older patients may deteriorate faster or present atypically, making age clinically relevant to triage risk.
8	Arrival mode	Arrival by ambulance or urgent transport may reflect higher acuity or greater need for rapid assessment.
9	Chest pain is the chief complaint	Chest pain can indicate time-sensitive cardiovascular emergencies.
10	Shortness of breath is the chief complaint	Shortness of breath may indicate respiratory distress, heart failure, pulmonary embolism or other urgent conditions.

E. Caveats

This dataset should be treated as suitable for exploration, not clinical deployment. Before modeling, missingness must be documented, implausible values must be reviewed, rare chief complaint indicators may need grouping and post-triage variables must be excluded to prevent data leakage.

Fairness and governance also matter. Race, ethnicity and insurance status should mainly support equity auditing and subgroup performance review rather than being used automatically as

clinical predictors. Any baseline model should be checked for performance differences across patient groups.

The dataset also reflects one hospital context. Therefore, any model developed from it should be described as a single-site baseline. Before wider use, it would require validation across different hospitals, patient populations, staffing patterns and documentation practices.

Final recommendation: Proceed to Week 6 baseline modelling as a controlled exploratory exercise, with clear documentation of missingness, target balance, clinical plausibility checks, data leakage exclusions and subgroup performance review.